AI Builder Application — Explanation

The Problem

Self-directed investing account opening is broken for a large segment of customers. The online form asks 100+ fields across KYC, suitability, tax residency, and compliance. Younger, financially literate users push through in 15–30 minutes. Everyone else — older customers, newcomers to Canada, anyone unfamiliar with financial jargon — hits a wall. "Are you a politically exposed person?" gets a blank stare. "Why do you need my SIN?" triggers distrust. Wrong account type gets selected because the form doesn't explain the difference. They abandon and call in. That call takes 40–60 minutes and costs the brokerage $25–40 in labor. The representative asks the exact same questions — the only difference is a human voice that can explain things.

The System

This project replaces both the form and the phone call with a conversational AI that collects every required field through natural dialogue. One question at a time, at the customer's pace, with explanations built into the flow.

The phone call succeeds because humans do three things forms can't: explain jargon in plain language, reassure on sensitive questions, and adapt to vague answers. This system does the same. "What's a TFSA?" gets a one-sentence answer, not a tooltip. "Why do you need my SIN?" gets "CRA requires it for registered accounts — it's encrypted, never shared." "I don't know, whatever's good" gets a follow-up question, not a validation error. The customer finishes the process without realizing they just completed a 100-field compliance form.

Where AI Stops

The AI collects and explains. It does not advise, assess suitability, or approve. Every completed application routes to a human for final review — the AI gathers, a person decides. Complex cases — dual citizens triggering FATCA, PEP disclosures, inconsistent information — get flagged for human review immediately rather than resolved by the model. Straightforward cases flow through end-to-end; anything carrying regulatory or reputational risk goes to a person with full context already assembled.

All boundaries are enforced through a plain-text rule file editable by compliance teams. Changes take effect on the next message — no code, no deployment, no release cycle.

What Breaks at Scale

Two things. First, prompt degradation — past 100+ rules, the model begins ignoring earlier instructions. The architecture addresses this with a hot-reloadable config injected server-side, allowing rules to be reorganized or split by context without application changes. Second, trust

erosion — one confidently wrong answer about contribution limits or tax obligations undermines every answer that follows. Every compliance rule is enforced as mandatory and auditable, not probabilistic.

Data Privacy

The system runs fully on-premises: open-source LLM via Ollama, Node.js server, zero external API calls. Customer SINs, financial profiles, and personal data never leave the company's network. Cloud models are available for internal testing; the production path is local and air-gapped.

Application to Wealthsimple

Account opening, KYC refresh, mortgage onboarding, tax filing intake — every product that funnels customers through a complex form has the same failure mode. The approach generalizes: conversational collection with hard compliance boundaries, human escalation for edge cases, and zero data leakage.